Progressive exercises

E1 · Practice. Solve the equation $\sin x = -\frac{1}{2}$ in $[-\pi, \pi]$.

E2 · Practice. Solve the inequality $\cos x \geq \frac{\sqrt{2}}{2}$ in $[-\pi, \pi]$.

E3 · Practice. Define on $\mathbb{R}$ the functions $f(x) = \cos(3x - 1)$ and $g(x) = \sin^2 x$. Calculate their derivatives.

E4 · Practice. Consider $f(x) = 2 + \sin x$ on $[0, 2\pi]$. Calculate its derivative and draw up its table of variations.

E5 · Practice. Calculate $\displaystyle\int_0^{\pi} \sin x \, \mathrm{d}x$, then the mean value of the sine function on $[0, \pi]$.

Textbook or past-paper work

Barbazo, p. 166, exercise 19. In S75: question 1, to read a symmetry and a period from a curve. At home: question 2, to prove the conjectures using identities valid for every real number.